Investment Gestation Lags, Intangible Capital, and the Measurement of Capital and Productivity in 39 Economies

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Abstract. Conventional growth accounting treats the investment-to-output lag as constant and negligible and omits intangible investment from produced capital. We jointly estimate a time-varying gestation lag and a country-specific intangible share for 39 economies over 1970–2019, reconciling production and wealth accounts. Correcting the lag raises log-level TFP and the implied labour share by a median 1.7 percentage points each; the intangible correction raises capital by up to 1.1 per cent. An observable, parameter-free tempo proxy corroborates the fitted lag.

Keywords: Solow residual; time-to-build; intangible capital; growth accounting; structural change; productivity measurement.

Word count (text + figure/table equivalents): approximately 11,400 (main text including references ≈ 9,100 words; 5 figures and 2 tables ≈ 2,250 word-equivalents at 250–500 per item).

Declarations

Competing interests. The author declares no competing interests.

Consent to participate. Not applicable.

Consent to publish. Not applicable.

Data and code availability. Penn World Table 10.01, World Bank CWON, World Bank WDI, and OECD GFCF data used in this study are publicly available from their respective providers. Frozen source extracts, checksums, analysis scripts, intermediate results, and manuscript sources are archived at https://github.com/bougtoir/gdp-tempo-paper.

Declaration of generative AI use. The author used generative AI to assist with formatting the text, choosing words that suited the tone, and writing analysis code. The author reviewed and edited the content as needed and takes full responsibility for the content of the published article.

Ethics approval. This research uses publicly available, aggregate macroeconomic data (Penn World Table 10.01, World Bank CWON, World Bank WDI, OECD GFCF). No human subjects, animals, or personally identifiable data were involved. Ethical approval was not required.

Funding. No external funding was received for this research.